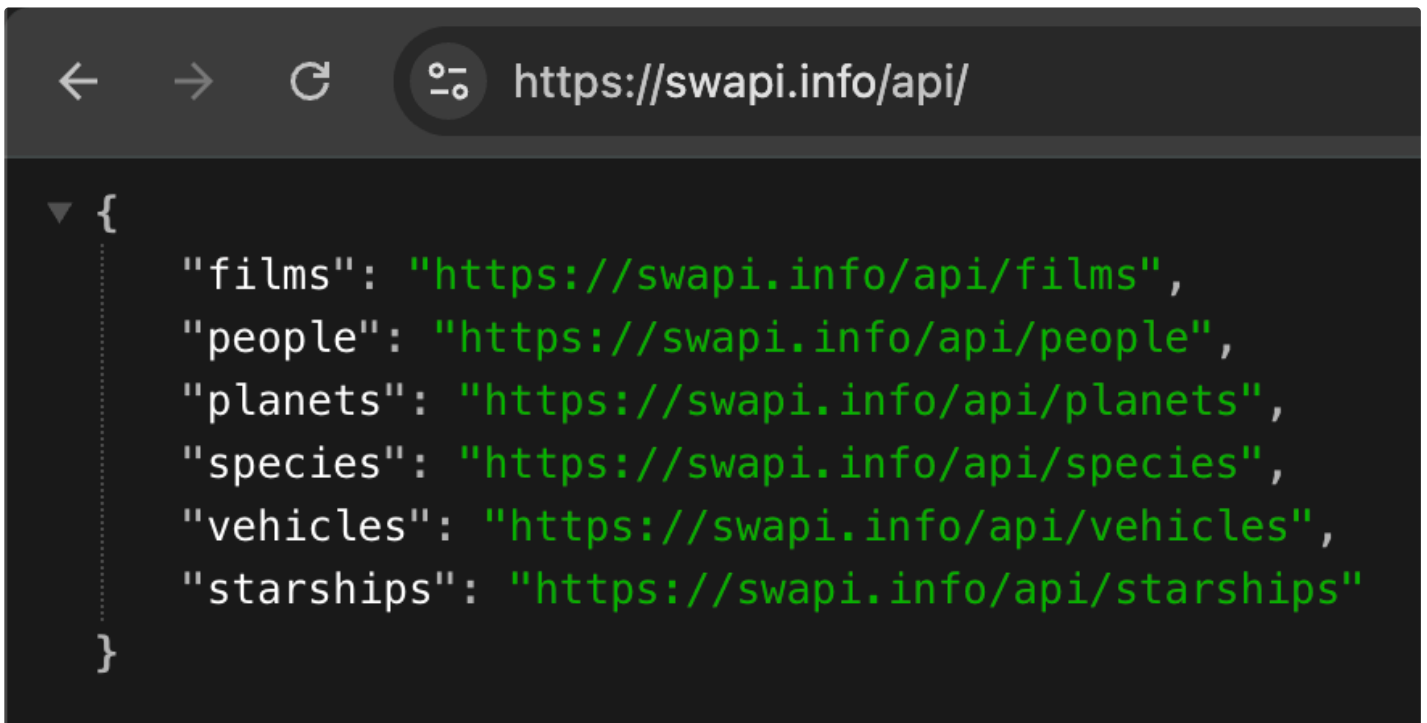


Assignment-5

Task1(a):

A screenshot of a web browser window. The address bar shows the URL "https://swapi.info/api/". Below the address bar, the JSON response is displayed in a dark-themed editor. The JSON is a root object with a single key "films" that points to another URL. The response is as follows:

```
{
  "films": "https://swapi.info/api/films",
  "people": "https://swapi.info/api/people",
  "planets": "https://swapi.info/api/planets",
  "species": "https://swapi.info/api/species",
  "vehicles": "https://swapi.info/api/vehicles",
  "starships": "https://swapi.info/api/starships"
}
```

When hitting the "<https://swapi.info/api/>" API endpoint, we get a JSON response containing a list of all the available API endpoints : films, people, planets, species, vehicles and starships.

1. The first endpoint "<https://swapi.info/api/films/>":

```
▼ [
  ▼ {
    "title": "A New Hope",
    "episode_id": 4,
    "opening_crawl": "It is a period of civil war.\r\nRebel spaceships, striking\r\nfrom a hidden base, have won\r\ntheir first victory against\r\nthe evil Galactic Empire.\r\n\r\nDuring the battle, Rebel\r\nspies managed to steal secret\r\nplans to the Empire's\r\nultimate weapon, the DEATH\r\nSTAR, an armored space\r\nstation with enough power\r\nto destroy an entire planet.\r\n\r\nPursued by the Empire's\r\nsinister agents, Princess\r\nLeia races home aboard her\r\nstarship, custodian of the\r\nstolen plans that can save her\r\npeople and restore\r\nfreedom to the galaxy....",
    "director": "George Lucas",
    "producer": "Gary Kurtz, Rick McCallum",
    "release_date": "1977-05-25",
    ▶ "characters": [ ... ], // 18 items
    ▶ "planets": [ ... ], // 3 items
    ▶ "starships": [ ... ], // 8 items
    ▶ "vehicles": [ ... ], // 4 items
    ▶ "species": [ ... ], // 5 items
    "created": "2014-12-10T14:23:31.880000Z",
    "edited": "2014-12-20T19:49:45.256000Z",
    "url": "https://swapi.info/api/films/1"
  },
  ▶ { ... }, // 14 items
  ▶ { ... }, // 14 items
  ▶ { ... }, // 14 items
  ▶ { ... }, // 14 items
  ▶ { ... } // 14 items
]
Raw Parsed
```

When calling the "films" endpoint we get a list of all the 6 movies, in each one of them there are scalar fields such as 'title' 'episode_id' and some reference fields, each one of them holds a list of API endpoints representing a relationship between the film and the field. As an example, the field 'characters' represents the 'people' that appears in that specific movie.

```
[
  {
    "title": "A New Hope",
    "episode_id": 4,
    "opening_crawl": "It is a period of civil war.\r\nRebel spaceships, striking\r\nfrom a hidden base, have won\r\ntheir first victory against\r\nthe evil Galactic Empire.\r\n\r\nDuring the battle, Rebel\r\nspies managed to steal secret\r\nplans to the Empire's\r\nultimate weapon, the DEATH\r\nSTAR, an armored space\r\nstation with enough power\r\nto destroy an entire planet.\r\n\r\nPursued by the Empire's\r\nsinister agents, Princess\r\nLeia races home aboard her\r\nstarship, custodian of the\r\nstolen plans that can save her\r\npeople and restore\r\nfreedom to the galaxy....",
    "director": "George Lucas",
    "producer": "Gary Kurtz, Rick McCallum",
    "release_date": "1977-05-25",
    "characters": [
      "https://swapi.info/api/people/1",
      "https://swapi.info/api/people/2",
      "https://swapi.info/api/people/3",
      "https://swapi.info/api/people/4",
      "https://swapi.info/api/people/5",
      "https://swapi.info/api/people/6",
      "https://swapi.info/api/people/7",
      "https://swapi.info/api/people/8",
      "https://swapi.info/api/people/9",
      "https://swapi.info/api/people/10",
      "https://swapi.info/api/people/12",
      "https://swapi.info/api/people/13",
      "https://swapi.info/api/people/14",
      "https://swapi.info/api/people/15",
      "https://swapi.info/api/people/16",
      "https://swapi.info/api/people/18",
      "https://swapi.info/api/people/19",
      "https://swapi.info/api/people/81"
    ],
    "planets": [ ... ], // 3 items
    "starships": [ ... ], // 8 items
    "vehicles": [ ... ], // 4 items
    "species": [ ... ], // 5 items
    "created": "2014-12-10T14:23:31.880000Z",
    "edited": "2014-12-20T19:49:45.256000Z",
    "url": "https://swapi.info/api/films/1"
  },
  { ... }, // 14 items
  { ... }, // 14 items
  { ... }, // 14 items
  { ... }, // 14 items
  { ... } // 14 items
]
```

When expanding the 'characters' field, we find each person in the film represented as an API endpoint with the specific id of that person, and in order to get the data of it, you need to call again that specific endpoint.

<u>scalar fields</u>	<u>reference fields</u>
-----------------------------	--------------------------------

- | | |
|--|---|
| <ul style="list-style-type: none">• title• episode_id• director• producer• release_date• created• edited• url | <ul style="list-style-type: none">• characters• planets• starships• vehicles• species |
|--|---|

We categorised 'url' as a scalar field not reference despite having it an API endpoint because it doesn't represent a relationship or connecting a different entity, it points to the object itself.

2. The API endpoint "<https://swapi.info/api/people>":

[illegible]

For the second endpoint we have the same as the first one, some scalar fields alongside some reference one to connect the data.

<u>scalar fields</u>	<u>reference fields</u>
• name • height • mass • hair_color • skin_color • eye_color • birth_year • gender • created • edited • url	• homeworld • films • species • vehicles • starships

3. The API endpoint : "<https://swapi.info/api/planets>":

```
▼ [
  ▼ {
    "name": "Tatooine",
    "rotation_period": "23",
    "orbital_period": "304",
    "diameter": "10465",
    "climate": "arid",
    "gravity": "1 standard",
    "terrain": "desert",
    "surface_water": "1",
    "population": "200000",
    "residents": [
      "https://swapi.info/api/people/1",
      "https://swapi.info/api/people/2",
      "https://swapi.info/api/people/4",
      "https://swapi.info/api/people/6",
      "https://swapi.info/api/people/7",
      "https://swapi.info/api/people/8",
      "https://swapi.info/api/people/9",
      "https://swapi.info/api/people/11",
      "https://swapi.info/api/people/43",
      "https://swapi.info/api/people/62"
    ],
    "films": [
      "https://swapi.info/api/films/1",
      "https://swapi.info/api/films/3",
      "https://swapi.info/api/films/4",
      "https://swapi.info/api/films/5",
      "https://swapi.info/api/films/6"
    ],
    "created": "2014-12-09T13:50:49.641000Z",
    "edited": "2014-12-20T20:58:18.411000Z",
    "url": "https://swapi.info/api/planets/1"
  },
  ▶ { ... }, // 14 items
  ▶ { ... }, // 14 items
  ▶ { ... }, // 14 items
  ▶ { ... }, // 14 items

```

<u>scalar fields</u>	<u>reference fields</u>
• name • rotation_period	• residents • films

- | | |
|---|--|
| <ul style="list-style-type: none">• orbital_period• diameter• climate• gravity• terrain• surface_water• population• created• edited• url | |
|---|--|

4. The API endpoint "<https://swapi.info/api/species>":

```

▼ [
  ▼ {
    "name": "Human",
    "classification": "mammal",
    "designation": "sentient",
    "average_height": "180",
    "skin_colors": "caucasian, black, asian, hispanic",
    "hair_colors": "blonde, brown, black, red",
    "eye_colors": "brown, blue, green, hazel, grey, amber",
    "average_lifespan": "120",
    "homeworld": "https://swapi.info/api/planets/9",
    "language": "Galactic Basic",
    ▼ "people": [
      "https://swapi.info/api/people/66",
      "https://swapi.info/api/people/67",
      "https://swapi.info/api/people/68",
      "https://swapi.info/api/people/74"
    ],
    ▼ "films": [
      "https://swapi.info/api/films/1",
      "https://swapi.info/api/films/2",
      "https://swapi.info/api/films/3",
      "https://swapi.info/api/films/4",
      "https://swapi.info/api/films/5",
      "https://swapi.info/api/films/6"
    ],
    "created": "2014-12-10T13:52:11.567000Z",
    "edited": "2014-12-20T21:36:42.136000Z",
    "url": "https://swapi.info/api/species/1"
  },
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items
  ▶ { ... }, // 15 items

```

Raw

Parsed

scalar fields

reference fields

- name
- classification
- designation
- average_height
- skin_colors
- hair_colors
- eye_colors
- average_lifespan
- language
- created
- edited
- url

- homeworld
- people
- films

5. The API endpoint "<https://swapi.info/api/vehicles>" :

```
▼ [
  ▼ {
    "name": "Sand Crawler",
    "model": "Digger Crawler",
    "manufacturer": "Corellia Mining Corporation",
    "cost_in_credits": "150000",
    "length": "36.8 ",
    "max_atmosphering_speed": "30",
    "crew": "46",
    "passengers": "30",
    "cargo_capacity": "50000",
    "consumables": "2 months",
    "vehicle_class": "wheeled",
    "pilots": [],
    ▼ "films": [
      "https://swapi.info/api/films/1",
      "https://swapi.info/api/films/5"
    ],
    "created": "2014-12-10T15:36:25.724000Z",
    "edited": "2014-12-20T21:30:21.661000Z",
    "url": "https://swapi.info/api/vehicles/4"
  },
  ▶ { ... }, // 16 items
  ▶ { ... }, // 16 items
  ▶ { ... }, // 16 items
  ▶ { ... }, // 16 items
  f ... // 16 items
]
```

<u>scalar fields</u>	<u>reference fields</u>
• name • model • manufacturer • cost_in_credits • length • max_atmosphering_speed	• pilots • films

- crew
- passengers
- cargo_capacity
- consumables
- vehicle_class
- created
- edited
- url

6. The API endpoint "<https://swapi.info/api/starships>":

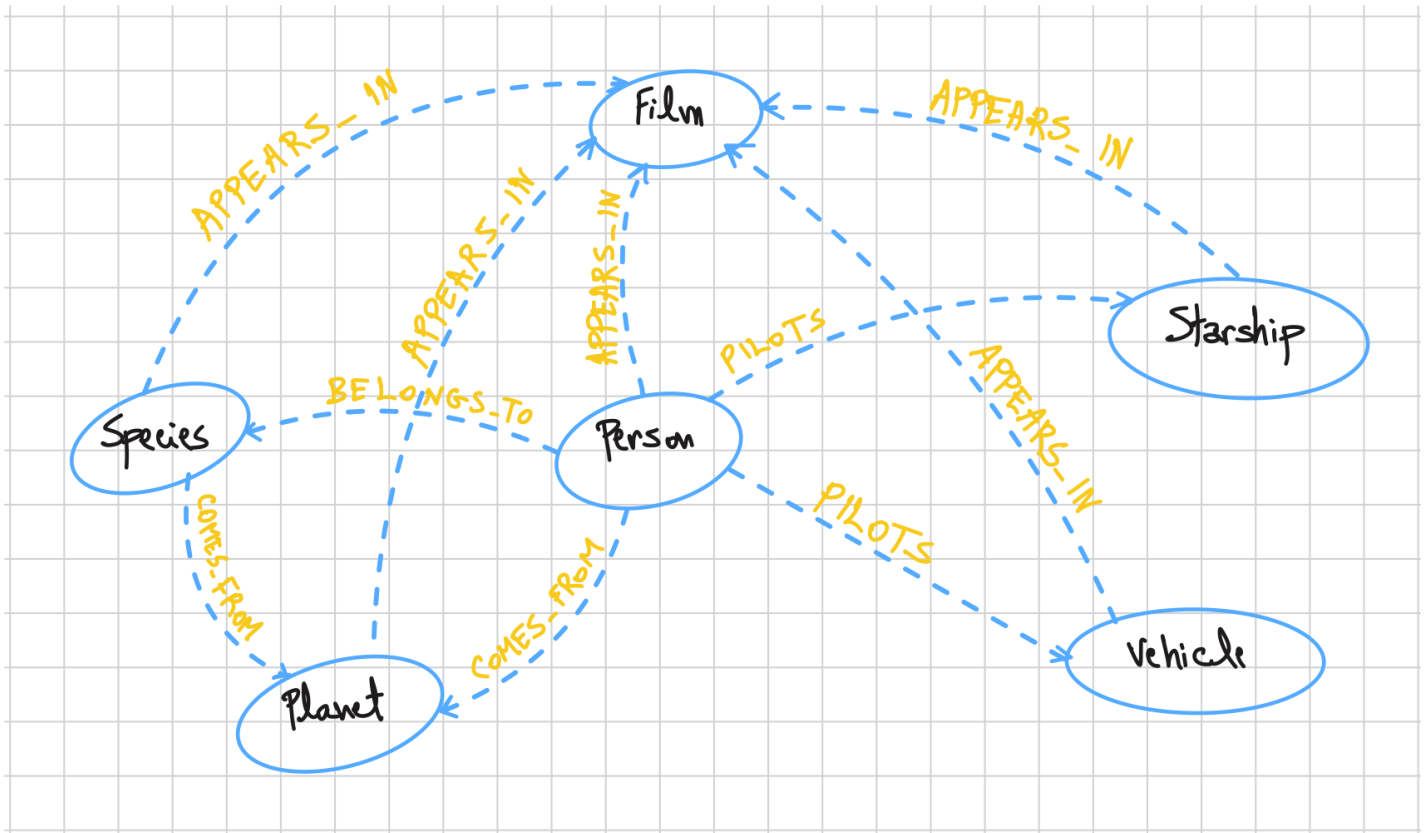
```
▼ [
  ▼ {
    "name": "CR90 corvette",
    "model": "CR90 corvette",
    "manufacturer": "Corellian Engineering Corporation",
    "cost_in_credits": "3500000",
    "length": "150",
    "max_atmosphering_speed": "950",
    "crew": "30-165",
    "passengers": "600",
    "cargo_capacity": "3000000",
    "consumables": "1 year",
    "hyperdrive_rating": "2.0",
    "MGLT": "60",
    "starship_class": "corvette",
    "pilots": [],
    ▼ "films": [
      "https://swapi.info/api/films/1",
      "https://swapi.info/api/films/3",
      "https://swapi.info/api/films/6"
    ],
    "created": "2014-12-10T14:20:33.369000Z",
    "edited": "2014-12-20T21:23:49.867000Z",
    "url": "https://swapi.info/api/starships/2"
  },
  ▶ { ... }, // 18 items
  ▶ { ... }, // 18 items
  ▶ { ... }, // 18 items
  ▶ { ... }, // 18 items

```

<u>scalar fields</u>	<u>reference fields</u>
• name • model • manufacturer • cost_in_credits • length	• pilots • films

- max_atmosphering_speed
- crew
- passengers
- cargo_capacity
- consumables
- hyperdrive_rating
- MGLT
- starship_class
- created
- edited
- url

Task1(b):



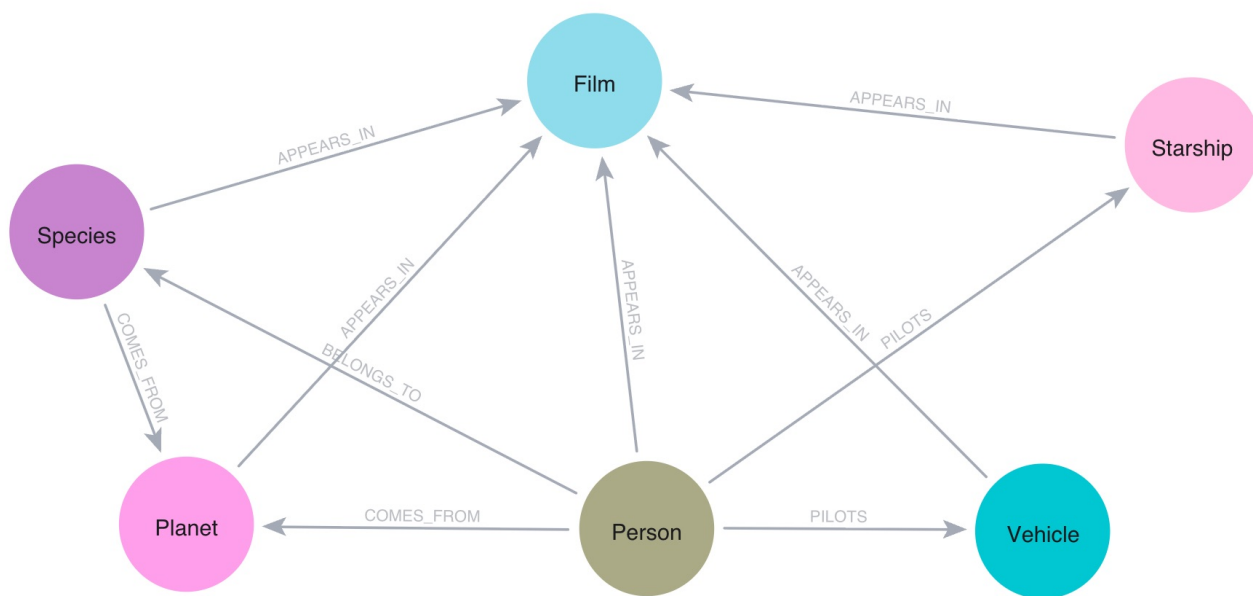
We chose Film, Species, Planet, Person, Starship and Vehicle as the graph nodes because they are basically the 6 API endpoints we have. We are going to keep all the scalar fields as node properties because they describe the characteristics of each node.

For the edges, we converted the reference fields to relationships between the nodes, we kept the naming simple and direct like a logical english sentence (Subject-verb-object), as well as we kept uniform relationship names to reduce schema complexity and make cypher queries easier.

We will take the url field, which we have in every object as an endpoint that has a unique identifier for that object, and make it as the node id. And we will convert all the numerical data from strings to numerical values in the importing phase.

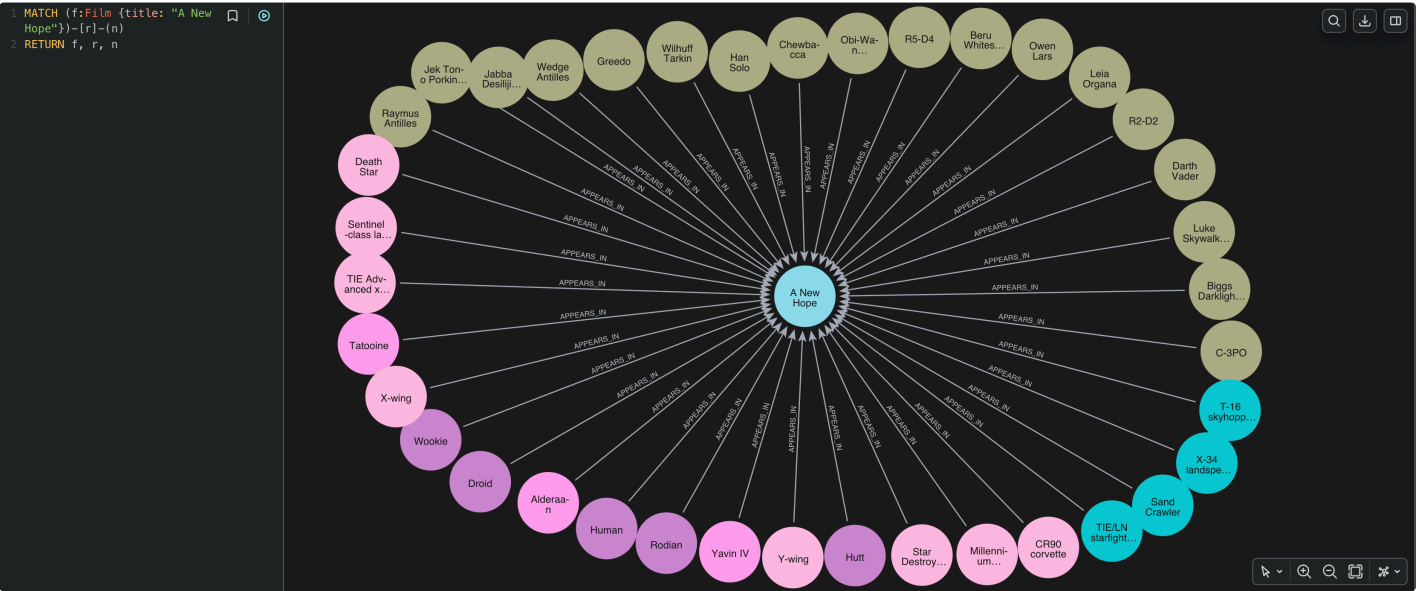
Task1(c):

After getting all the data from the 6 API endpoints and importing it in neo4j graph, we get the following schema that matches perfectly the one we drew before.



We made also a query to get 'APPEARES_IN'

relationship for the movie : " A new Hope".



The full python code with comments is in the file "code.py"